Technical University of Crete
School of Electrical and Computer Engineering
Course: Optimization
Exercise 3 (100/1000)
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In this exercise we shall solve simple, but very important, optimization problems by explicitly solving the KKT conditions.

1. In the first exercise, we were asked to compute the projection of $x_0 \in \mathbb{R}^n$ onto the set $\mathbf{B}(\mathbf{0}, r) := \{\mathbf{x} \in \mathbb{R}^n | \parallel x \parallel_2 \leq \ r\}$.

(a) We started by drawing a scheme of the problem with simple Matlab code. The produced plot is below:

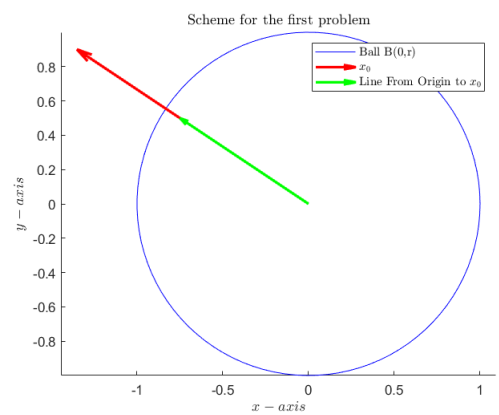

Figure 1: Scheme for the first exercise

(b) Then we write our optimization problem in terms of differentiable functions as follows

$$minimize \ \ f_0(x) = \parallel x_0 - x \parallel_2^2$$

$$subject \ \ to \ \ f_i(x) = \parallel x_i \parallel_2^2 - r^2 \leq 0, \ \ i = 1, .., n$$

(c) The Karush-Kuhn-Tucker (KKT) conditions are:

$$\nabla f_0(x_*) + \sum_{i=1}^{n} \lambda_i \nabla f_i(x_*) = 0 \tag{1}$$

$$\lambda_i \geq 0, \ \ for \ \ i = 1, ...n \tag{2}$$

$$f_i(x_*) \leq 0, \ \ for \ \ i = 1, ...n \tag{3}$$

$$\lambda_i f_i(x_*) = 0, \ \ for \ \ i = 1, ...n \tag{4}$$

(d) We consider the case where $\lambda_* \geq 0$. Then we take the first of the KKT conditions and we have that

$$\nabla f_0(x_*) + \sum_{i=1}^{n} \lambda_i \nabla f_i(x_*) = 0 \Rightarrow$$

$$-2(x_0 - x_*) + \lambda_* 2x_* = 0 \Rightarrow$$
$$-2x_0 + 2x_* + \lambda 2x_* = 0 \Rightarrow$$
$$2x_*(\lambda_* + 1) = 2x_0$$
$$x_* = \frac{x_0}{\lambda_* + 1} \tag{5}$$

For the third KKT condition we have : $\lambda_*(\| x_* \|_2^2 - r^2) = 0$.
Because of $\lambda_* \geq 0$ we have that, in order to satisfy the relationship we should:

$$\| x_* \|_2^2 = r^2 \Rightarrow$$
$$\| \frac{x_0}{\lambda_* + 1} \|_2^2 = r^2 \Rightarrow$$

because both of the quantities are positive we can write that

$$\frac{\| x_0 \|}{\lambda_* + 1} = r \Rightarrow$$
$$\lambda_* r + r = \| x_0 \| \Rightarrow$$
$$\lambda_* = \frac{\| x_0 \|}{r} - 1$$

We substitute the above quantity in relationship (5) so we get that

$$x_* = \frac{x_0}{\frac{\|x_0\|}{r} - 1 + 1} \Rightarrow$$
$$x_* = r\frac{x_0}{\| x_0 \|} = P_B(x_0)$$

What we observe is that the projection is a vector scaled with r. From the projection we derive that projecting x_0, even if it is not inside the ball, the projection is onto the boundary of the ball. If $\| x_0 \|_2 \leq r$, then the point itself is the projection onto the set $\mathbf{B}(\mathbf{0}, r)$.
Actually, that is the case we will examine now.

(e) Moving on, we will assume that $\lambda_* = 0$. From the first condition we have:

$$\nabla f_0(x) = 0 \Rightarrow$$
$$-2(x_0 - x_*) = 0 \Rightarrow$$
$$x_* = x_0$$

The assumption that $\lambda_* = 0$ indicates that $f_i(x_*)$ is an inactive constraint, because the corresponding Lagrange multiplier is zero ans also we understand that $x_0 \in \mathbf{B}(\mathbf{0}, r)$ and that is the reason why the projection is x_0 itself.

2. In this exercise we are asked to compute the projection of $x_0 \in \mathbb{R}^n$ onto the set $\mathbf{B}(\mathbf{y}, r) := \{\mathbf{x} \in \mathbb{R}^n | \parallel x - y \parallel_2 \leq r\}$. We will follow all the previous steps. First, we drew a scheme of our problem with simple Matlab code.

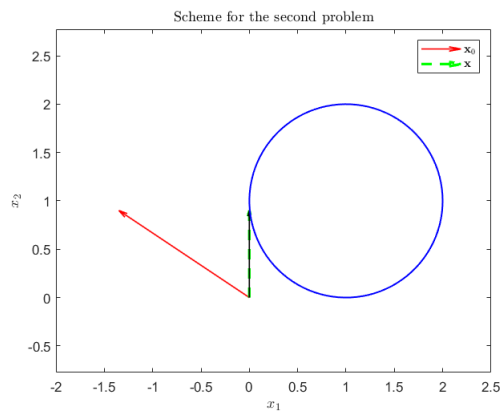

Figure 2: Scheme for the second problem

Then, we went on to write our problem in a form with differentiable functions.

$$minimize \;\; f_0(x) = \parallel x_0 - x \parallel_2^2$$

$$subject \;\; to \;\; f_i(x) = \parallel x_i - y \parallel_2^2 - r^2 \leq 0, \;\; i = 1, .., n$$

The KKT conditions for this problem are the following:

$$\nabla f_0(x_*) + \sum_{i=1}^{n} \lambda_i \nabla f_i(x_*) = 0 \tag{6}$$

$$\lambda_i \geq 0, \;\; for \;\; i = 1, ...n \tag{7}$$

$$f_i(x_*) \leq 0, \;\; for \;\; i = 1, ...n \tag{8}$$

$$\lambda_i f_i(x_*) = 0, \;\; for \;\; i = 1, ...n \tag{9}$$

Then, we consider $\lambda_i > 0$. From the first KKT condition we have that

$$\nabla f_0(x_*) + \sum_{i=1}^{n} \lambda_* \nabla f_i(x_*) = 0 \Rightarrow$$

$$-2(x_0 - x_*) + \lambda_* 2(x_* - y) = 0 \Rightarrow$$

$$-2x_0 + 2x_* + 2\lambda_* x_* - 2\lambda_* y = 0 \Rightarrow$$

$$x_*(2 + 2\lambda_*) = 2(y\lambda_* + x_0) \Rightarrow$$

$$x_* = \frac{x_0 + y\lambda_*}{1 + \lambda_*}$$

Then, we proceed to substitute the x_* from above in the third KKT condition. We have that,

$$\lambda_i(f_i(x_*)) = 0 \Rightarrow$$

and because $\lambda_* > 0$

$$\| x_* - y \|_2^2 - r^2 = 0 \Rightarrow$$

$$\| \frac{x_0 + y\lambda_*}{1 + \lambda_*} - y \|_2^2 - r^2 = 0 \Rightarrow$$

$$\| \frac{x_0 - y}{1 + \lambda_*} \|_2^2 = r^2 \Rightarrow$$

$$\frac{\| x_0 - y \|}{1 + \lambda_*} = r \Rightarrow$$

$$\lambda_* = \frac{\| x_0 - y \|}{r} - 1$$

Then, we take the λ_* and we substitute in the relation we have for x_*. That gives us:

$$x_* = y + \frac{r}{\| x_0 - y \|}(x_0 - y)$$

Moving forwards, we considered that $\lambda_* = 0$. The first KKT condition gives us

$$\nabla f_0(x_*) = 0 \Rightarrow$$

$$-2(x_0 - x_*) = 0 \Rightarrow$$

$$x_* = x_0$$

As in the first exercise, we derive the conclusion that when the point belongs in the ball the projection is the point itself.

3. In the third exercise, once again, we are asked to compute the projection of $x_0 \in \mathbb{R}^n$ onto the set $\mathbf{B}(\mathbf{y}, r) := \{\mathbf{x} \in \mathbb{R}^n | \mathbf{a} \leq \mathbf{x}\}$. First, we write our problem in the form of:

$$minimize \;\; f_0(x) = \frac{1}{2} \parallel x_0 - x \parallel_2^2$$

$$subject \;\; to \;\; f_i(x) = \mathbf{a} - \mathbf{x_i} 0, \;\; i = 1, .., n$$

Then, we proceed to write the KKT conditions as follows:

$$\nabla f_0(x_*) + \sum_{i=1}^{n} \lambda_i \nabla f_i(x_*) = 0$$

$$\lambda_i \geq 0, \;\; for \;\; i = 1, ...n$$

$$f_i(x_*) \leq 0, \;\; for \;\; i = 1, ...n$$

$$\lambda_i f_i(x_*) = 0, \;\; for \;\; i = 1, ...n$$

And then we move on to compute:

$$\nabla f_0(x_*) = -\frac{1}{2} 2(x_0 - x_*) = x_* - x_0$$

and

$$\nabla f_i(x_*) = -1$$

So the first KKT condition is transformed in

$$x_* - x_0 - \lambda_* = 0 \Rightarrow$$

$$x_* = \lambda_* + x_0 \geq 0$$

Now, we will examine the cases where $\lambda_* = 0$ where

$$x_* = x_0, \;\; x_0 \in \mathbf{S}$$

Once again, we observe that when x_0 is a point that belongs in the examined set, the projection is the point itself. Now, we will examine the case where $\lambda_* > 0$, that means that $f_i(x_*)$ is an active constraint.

$$\lambda_* = -x_0 + a$$

so

$$x_* = -x_0 + a + x_0 \Rightarrow$$

$$x_* = a$$

So, $\mathbb{P}(x_0) = max\{a, x_0\}$.

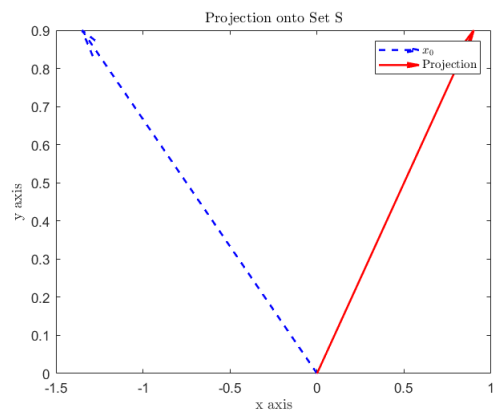

Figure 3: Scheme for exercise 3

4. The current optimization problem in exercise 4 is to

$$minimize \;\; \frac{1}{2} \parallel x_0 - x \parallel_2^2$$

$$subject \;\; to \;\; \mathbf{Ax} = \mathbf{b}$$

The KKT conditions are:

$$\nabla f_0(x_*) + \mathbf{v} \nabla f_i(x_*) = 0 \Rightarrow$$

$$\nabla f_0(x_*) + \mathbf{A^T v} = 0 \Rightarrow$$

$$x_* = x_0 - \mathbf{A^T v}$$

And we also have the condition that

$$\mathbf{A} x_* = b$$

This kind of equations can be written as

$$\begin{bmatrix} \mathbb{I}_{\ltimes \times \ltimes} & \mathbf{A^T} \\ \mathbf{A} & \mathbf{0} \end{bmatrix} \begin{bmatrix} x_* \\ \mathbf{v} \end{bmatrix} = \begin{bmatrix} x_0 \\ b \end{bmatrix}$$

The matrix $\begin{bmatrix} \mathbb{I}_{\ltimes \times \ltimes} & \mathbf{A^T} \\ \mathbf{A} & \mathbf{0} \end{bmatrix}$ has dimensions of $(n+m) \times (n+m)$ so it is invertible. That meanes that the solution of the system can be written as

$$\begin{bmatrix} x_* \\ \mathbf{v} \end{bmatrix} = \begin{bmatrix} \mathbb{I}_{\ltimes \times \ltimes} & \mathbf{A^T} \\ \mathbf{A} & \mathbf{0} \end{bmatrix}^{-1} \begin{bmatrix} x_0 \\ b \end{bmatrix}$$

So the system has unique solution the optimal pair (x_*, v_*) with x_* being $x_* = \begin{bmatrix} x_0 \\ b \end{bmatrix}$.

5. In this exercise we are given the set $S := \{x \in \mathbb{R}^2 \mid \|x\|_2 \leq 1 \ , \ x_1 \geq 0 \ , \ x_2 \geq 0 \ , \ x_1 + x_2 \leq 1\}$ and we are asked to minimize the function $f_0(x)$. First, we draw the set so we can check if we have any redundant constraints.

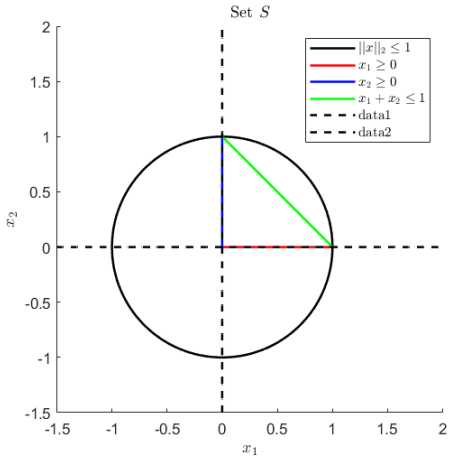

Figure 4: The set $\mathbb{S}$

From the sketch we can draw the conclusion that the constraint $\|x\|_2 \leq 1$ is a redundant constraint. So, in fact we have the set $S := \{x \in \mathbb{R}^2 \mid x_1 \geq 0 \ , \ x_2 \geq 0 \ , \ x_1 + x_2 \leq 1\}$

(a) We have the function

$$f_0(x) = (x_1 - 2)^2 + (x_2 - 2)^2$$

When we have that kind of problems we write the inequalities in forms we can handle. So, we end up having to solve the optimization problem

$$minimize \ \ f_0(x) = (x_1 - 2)^2 + (x_2 - 2)^2$$

$$subject \ \ to \ \ f_1(x) = -x_1 \leq 0$$

$$f_2(x) = -x_2 \leq 0$$

$$f_3(x) = x_1 + x_2 - 1 \leq 0$$

The KKT conditions are:

$$\nabla f_0(x_*) + \sum_{i=1}^{n} \lambda_i \nabla f_i(x_*) = 0$$

$$\lambda_i \geq 0, \quad for \ \ i = 1, 2, 3$$

$$f_i(x_*) \leq 0, \quad for \ \ i = 1, 2, 3$$

$$\lambda_i f_i(x_*) = 0, \quad for \;\; i = 1, 2, 3$$

We compute that

$$\nabla f_0(x) = \begin{bmatrix} \frac{\partial f_0}{\partial x_1} \\ \frac{\partial f_0}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 2x_1 - 4 \\ 2x_2 - 4 \end{bmatrix}$$

Also, we compute that: $\nabla f_1(x) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \nabla f_2(x) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \nabla f_3(x) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Now, we proceed to substitute the above relationships into the first KKT condition:

$$\begin{bmatrix} 2x_1 - 4 \\ 2x_2 - 4 \end{bmatrix} + \lambda_1 \begin{bmatrix} -1 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 0 \\ -1 \end{bmatrix} + \lambda_3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 0 \Rightarrow$$

$$\begin{bmatrix} 2x_1 - 4 \\ 2x_2 - 4 \end{bmatrix} + \begin{bmatrix} -\lambda_1 + \lambda_3 \\ -\lambda_2 + \lambda_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

So we have that

$$2x_1 - 4 - \lambda_1 + \lambda_3 = 0$$

$$2x_2 - 4 - \lambda_2 + \lambda_3 = 0$$

So x_1, x_2 are

$$x_1 = \frac{\lambda_1 - \lambda_3 + 4}{2}$$

$$x_2 = \frac{\lambda_2 - \lambda_3 + 4}{2}$$

From the fourth KKT condition we have that $\lambda_1 x_1 = 0$ and $\lambda_2 x_2 = 0$ so we distinct cases for λ_i. For the optimal point x_* we start with the case that $f_3(x) < 0$ so it is an inactive constraint.

1. First case we have that $\lambda_3 = 0$ so $x_1 = \frac{\lambda_1 + 4}{2}$ and $x_2 = \frac{\lambda_2 + 4}{2}$.
   Then we distinct the following cases for λ_1 and λ_2.

   - $\lambda_1 = 0$ and $\lambda_2 = 0$
     That gives us $x_1 = 2$ and $x_2 = 2$. It also means that all the constraints are inactive. But if $x_1 = 2$ and $x_2 = 2$ then $f_3(x) < 0 \Rightarrow x_1 + x_2 - 1 < 0$ and that turns out to be absurd.
   - $\lambda_1 = 0$ and $\lambda_2 > 0$
     That gives us $x_1 = 2$ and $x_2 = \frac{\lambda_2 + 4}{2} > 0$. Again we examine the relation

$$f_3(x) < 0 \Rightarrow x_1 + x_2 - 1 < 0 \Rightarrow$$

$$2 + \lambda_2 + 4 < 0 \Rightarrow$$

$$\lambda_2 < -6$$

which is a contradiction to the hypothesis that $\lambda_2 > 0$. So that case is also an absurd.

- $\lambda_1 > 0$ and $\lambda_2 = 0$
  As above we end up in an absurd case.

- $\lambda_1 > 0$ and $\lambda_2 > 0$
  Now we have that $x_1 = \frac{\lambda_1+4}{2}$ and $x_2 = \frac{\lambda_2+4}{2}$ both positive quantities. We take the relation that $f_3(x) < 0$ and we end up

$$\frac{\lambda_1}{2} + 2 + \frac{\lambda_2}{2} + 2 - 1 < 0 \Rightarrow$$

$$\frac{\lambda_1}{2} + \frac{\lambda_2}{2} + 3 < 0$$

  which once again is absurd.

2. So, we are led to $\lambda_3 > 0$ and $f_3(x) = 0$, with that meaning that $f_3(x)$ is an active constraint and that the optimal point is inside the set $\mathbb{S}$. With that being said, we have that $x_1 = \frac{\lambda_1-\lambda_3+4}{2}$ and $x_2 = \frac{\lambda_2-\lambda_3+4}{2}$.

   - $\lambda_1 = 0$ and $\lambda_2 = 0$
     Now, that gives us $x_1 = \frac{4-\lambda_3}{2}$ and $x_2 = \frac{4-\lambda_3}{2}$.

$$f_3(x) = 0 \Rightarrow x_1 + x_2 - 1 = 0 \Rightarrow$$

$$2\frac{4-\lambda_3}{2} - 1 = 0 \Rightarrow \lambda_3 = 3$$

     So, $x_1 = \frac{1}{2}$ and $x_2 = \frac{1}{2}$ and, finally, $x_* = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$.

   - $\lambda_1 > 0$ and $\lambda_2 > 0$
     That means $f_1(x)$ and $f_2(x)$ active constraints so, $f_1(x) = 0$ and $f_2(x) = 0$ which leads to $x_1 = 0$ and $x_2 = 0$. From $f_3(x) = 0 \Rightarrow x_1 + x_2 - 1 = 0$ we end up in something absurd.

   - $\lambda_1 > 0$ and $\lambda_2 = 0$
     Now, $x_1 = \frac{\lambda_1-\lambda_3+4}{2} = 0$, $x_2 = \frac{4-\lambda_3}{2}$. From the equation where $f_3(x) = 0$ we have that

$$x_1 + x_2 - 1 = 0 \Rightarrow x_2 = 1 \Rightarrow \lambda_3 = 2$$

     For $\lambda_3 = 2$:

$$\lambda_1 = -2$$

     which is absurd.

- $\lambda_1 = 0$ and $\lambda_2 < 0$
  From that we have $x_2 = \frac{\lambda_1 - \lambda_3 + 4}{2} = 0$, $x_1 = \frac{4 - \lambda_3}{2}$ and as above we end up in absurd.
  So for $f_0(x) = (x_1 - 2)^2 + (x_2 - 2)^2$ the optimal point $x_* = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$.

(b) For this part

$$f_0(x) = (x_1 + 2)^2 + (x_2 + 2)^2$$

. We have the same KKT conditions and the optimization problem has the same form as above. Now,

$$\nabla f_0(x) = \begin{bmatrix} \frac{\partial f_0}{\partial x_1} \\ \frac{\partial f_0}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 2x_1 + 4 \\ 2x_2 + 4 \end{bmatrix}$$

and we also compute that $\nabla f_1(x) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \nabla f_2(x) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \nabla f_3(x) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Therefore, the first KKT condition has the form of

$$\begin{bmatrix} 2x_1 + 4 \\ 2x_2 + 4 \end{bmatrix} + \lambda_1 \begin{bmatrix} -1 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 0 \\ -1 \end{bmatrix} + \lambda_3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 0 \Rightarrow$$

$$\begin{bmatrix} 2x_1 + 4 \\ 2x_2 + 4 \end{bmatrix} + \begin{bmatrix} -\lambda_1 + \lambda_3 \\ -\lambda_2 + \lambda_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

So we have that

$$2x_1 + 4 - \lambda_1 + \lambda_3 = 0$$

$$2x_2 + 4 - \lambda_2 + \lambda_3 = 0$$

So x_1, x_2 are

$$x_1 = \frac{\lambda_1 - \lambda_3 - 4}{2}$$

$$x_2 = \frac{\lambda_2 - \lambda_3 - 4}{2}$$

We will follow the same procedure we did before.

1. We assume that $\lambda_3 = 0$ and so, $x_1 = \frac{\lambda_1 - 4}{2}$ and $x_2 = \frac{\lambda_2 - 4}{2}$. So, now we go on to distinct the cases for λ_1 and λ_2.

   - $\lambda_1 = 0$ and $\lambda_2 = 0$
     With that hypothesis we derive $x_1 = -2$ and $x_2 = -2$. But we have assumed that $\lambda_1 = 0$ and λ_2 which means that those constraints are inactive therefore $f_1(x) < 0 \Rightarrow x1 > 0$ and $f_2(x) < 0 \Rightarrow x2 > 0$. So we ended in an absurd hypothesis.

- $\lambda_1 > 0$ and $\lambda_2 = 0$
  which leads us to $x_1 = \frac{\lambda_1 - 4}{2}$ and $x_2 = -2$. But $f_2(x) < 0$ and $x_2 > 0$ so, again we ended up in an contradiction, that indicates that the hypothesis is absurd.
- $\lambda_1 = 0$ and $\lambda_2 < 0$
  for the same reason this is also an absurd hypothesis.
- $\lambda_1 > 0$ and $\lambda_2 > 0$
  In that case, we have that $x_1 = 0$ and $x_2 = 0$ so $\lambda_1 = 4$ and $\lambda_2 = 4$. We observe that all the constraints are satisfied, so we have an optimal point $x_* = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

We will continue to examine all the cases, but below we expect that all the assumptions will be absurd because the optimal point has been found.

2. Now we will assume that $\lambda_3 > 0$ so $f_3(x) = 0$.

   - $\lambda_1 = 0$ and $\lambda_2 = 0$
     In that case, $x_1 = \frac{-\lambda_3 - 4}{2}$ and $x_2 = \frac{-\lambda_3 - 4}{2}$. We substitute in the KKT $\lambda_3 f_3(x) = 0$ so
     $$f_3(x) = 0 \Rightarrow x_1 + x_2 - 1 = 0 \Rightarrow \lambda_3 = -4$$
     which, as expected, is an absurd result.
   - $\lambda_1 > 0$ and $\lambda_2 = 0$
     Now, we get that $x_1 = \frac{\lambda_1 - \lambda_3 - 4}{2}$ and $x_2 = \frac{-\lambda_3 - 4}{2}$. If $\lambda_1 > 0$ that gives us that $f_1(x) = 0$ as an active constraint. From that we get $x_1 = 0$. We substitute in
     $$f_3(x) = 0 \Rightarrow x_1 + x_2 - 1 = 0 \Rightarrow x_2 = 1 \Rightarrow$$
     $$\frac{-\lambda_3 - 4}{2} = 1 \Rightarrow \lambda_3 = -6$$
     which is absurd.
   - $\lambda_1 = 0$ and $\lambda_2 > 0$
     For the same reason as above we end up in an absurd result.
   - $\lambda_1 > 0$ and $\lambda_2 > 0$
     From that assumption we have that $x_1 = 0$ and $x_2 = 0$ and if we substitute them in the KKT $\lambda_3 f_3(x) = 0$ where $\lambda_3 > 0$ we get that $-1 = 0$ which is definetely absurd.

So, for $f_0(x) = (x_1 + 2)^2 + (x_2 + 2)^2$ the optimal point $x_* = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.